

SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 2-COMPARATIVE ANALYSIS

Course Code:	CSA1223	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2-COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
Student Name:	M. Nivedha	Reg.No.:	192471040
Department:	CS & BS	Date:	

ANNEXURE A COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I M. Nivedha (192471040) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1	0.5	1	1	0.5	0.75
Q2	0.5	1	1	0.5	0.5
Q3	1	1.25	0.5	1	0.5
Q4	1	1	0.5	1	0.75
Q5	0.5	1.25	0.5	0.5	0.5

RUBRICS FOR CALCULATION:

19

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ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

Compare L1, L2, and L3 Cache Levels in Terms of Latency, Bandwidth, and Throughput

Introduction

Cache memory is a small, high-speed memory located close to the processor. Modern processors generally use three cache levels: **L1, L2, and L3**. These cache levels reduce the time required to access frequently used data and instructions.

1. **L1 cache** is the smallest and fastest cache and is located closest to the CPU core.
2. **L2 cache** provides a balance between speed and capacity and acts as a backup when L1 does not contain the required data.
3. **L3 cache** is generally the largest cache and is often shared between multiple CPU cores.
4. The memory hierarchy is: **L1 → L2 → L3 → DRAM**.
5. As we move down the hierarchy, **capacity increases**, while **access speed decreases and latency increases**.
6. A high **cache-hit rate** reduces CPU waiting time and improves processor throughput.
7. Multiple cache levels balance **speed, capacity, bandwidth, power consumption, and cost**.

Comparison

Parameter	L1 Cache	L2 Cache	L3 Cache
Size	Smallest	Medium	Largest
Latency	Lowest	Low	Higher
Bandwidth	Very High	Very High	High
Throughput	Very High	High	High
Access Speed	Fastest	Very Fast	Relatively Slower
Location	Closest to CPU Core	Near CPU Core	Usually Shared

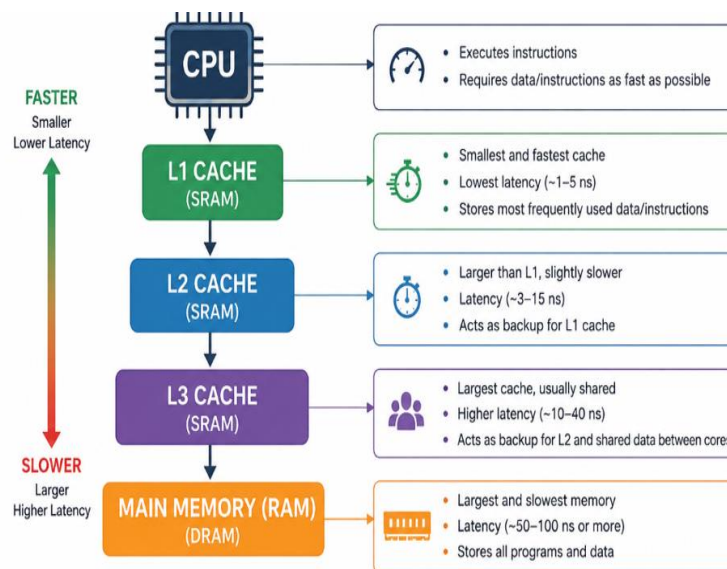


Fig: CPU Cache Memory Hierarchy – L1, L2, L3 Cache and Main Memory

Latency

L1 cache has the **lowest latency** because it is located closest to the processor and is designed for extremely fast access.

L2 cache has slightly higher latency because it provides greater storage capacity. L3 cache is generally larger and often shared between processor cores, so its access latency is higher than L1 and L2.

The hierarchy can be represented as:

L1 → L2 → L3 → DRAM

As we move down the memory hierarchy, capacity generally increases while access speed decreases.

Bandwidth and Throughput

L1 cache provides very high bandwidth and allows the CPU to quickly access frequently used data. L2 cache also provides high bandwidth and supplies data when an L1 cache miss occurs.

L3 cache provides a larger shared storage area and high aggregate bandwidth, although its access latency is greater than L1 and L2.

A **high cache-hit rate** improves processor throughput because the CPU spends less time waiting for data from slower DRAM main memory.

Evidence

Representative cache latency values are approximately **1–4 CPU cycles for L1, 5–15 CPU cycles for L2, and 20–50+ CPU cycles for L3**, depending on the processor architecture. These values demonstrate that **L1 provides the fastest access**, while L3 provides greater capacity with higher latency. A higher cache-hit rate reduces accesses to slower DRAM and can improve overall processor throughput.

Critical Analysis

A larger cache is not always better. Increasing cache size can improve the cache-hit rate, but it may also increase:

- Access latency
- Chip area
- Power consumption
- Manufacturing cost

Therefore, modern processors use multiple cache levels to achieve a balance between **speed, capacity, bandwidth, and cost**.

Conclusion

L1 cache provides the fastest access and lowest latency. L2 cache provides a balance between speed and capacity, while L3 cache provides larger shared cache capacity.

Together, L1, L2, and L3 caches reduce memory-access latency, improve effective bandwidth, increase processor throughput, and enhance overall processor performance.

2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

Differentiate SRAM and DRAM by Evaluating Their Latency, Bandwidth, and Throughput Characteristics in Cache and Main Memory Design

Introduction

SRAM and DRAM are two important types of semiconductor memory used in computer systems. Both are **volatile memories**, meaning that they lose stored data when power is removed. However, they differ significantly in terms of **latency, bandwidth, throughput, capacity, and cost**.

1. **SRAM** provides very low latency and is mainly used for **L1, L2, and L3 cache**.
2. **DRAM** provides larger capacity and is mainly used as **main memory**.
3. SRAM is faster than DRAM but is more expensive and has lower storage density.
4. DRAM has higher latency but provides high capacity at a lower cost per bit.
5. SRAM and DRAM work together in the memory hierarchy to balance **speed, capacity, throughput, and cost**.

Comparison

Parameter	SRAM	DRAM
Full Form	Static Random-Access Memory	Dynamic Random-Access Memory
Storage Method	Flip-flop based	Capacitor based
Latency	Very low	Higher
Bandwidth	Very high	High
Throughput	Very high	High
Refresh	Not required	Required
Density	Lower	Higher
Cost per Bit	Higher	Lower
Typical Use	L1, L2, L3 Cache	Main Memory
Capacity	Smaller	Larger

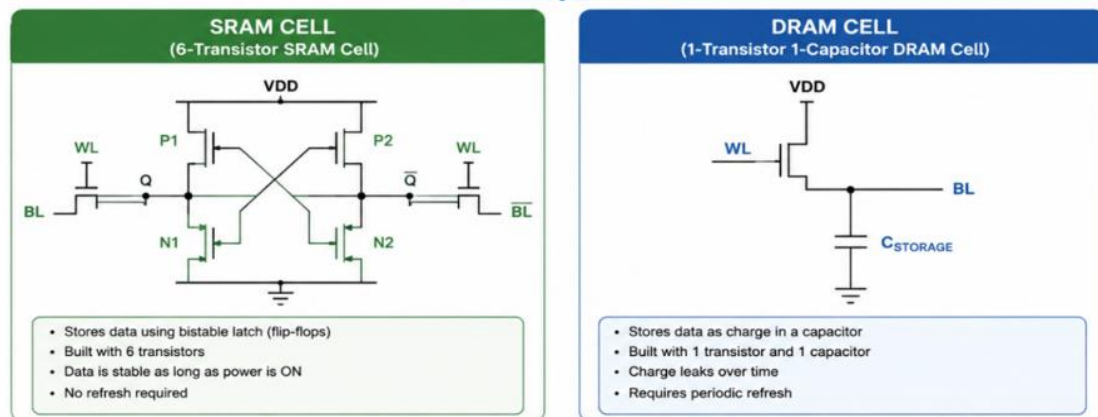


Fig: Comparison of SRAM and DRAM Memory Cells

Latency

SRAM has **lower latency** and can provide very fast access to stored data. This makes it suitable for CPU cache memory, where even a small delay can affect processor performance.

DRAM has higher latency compared with SRAM. However, it provides much greater storage capacity, making it suitable for main memory.

Bandwidth and Throughput

SRAM provides **very high bandwidth and throughput** because it is used close to the processor and can quickly supply frequently accessed data.

DRAM also provides high bandwidth through technologies such as:

- Multiple memory channels
- Memory banks
- Burst data transfers

However, its individual memory-access latency is generally higher than SRAM.

Use in Cache and Main Memory

The typical memory arrangement is:

CPU → SRAM Cache → DRAM Main Memory

SRAM is used for cache because **speed and low latency** are the main requirements.

DRAM is used as main memory because **large capacity and lower cost per bit** are more important.

Evidence

SRAM is commonly used for **CPU cache memory** because it provides very low access latency and fast data delivery. Representative SRAM cache access latency is typically only a **few CPU cycles**, whereas DRAM main-memory access may require **tens to hundreds of CPU cycles**, depending on the processor and memory architecture.

DRAM is used as **main memory** because it provides much greater storage capacity at a lower cost per bit. For example, modern processors use **SRAM-based L1, L2, and L3 caches**, while system main memory is typically implemented using **DRAM**. This demonstrates the practical trade-off between **speed, capacity, and cost**.

Critical Analysis

Using SRAM as the entire main memory would provide excellent speed, but it would be expensive and require a large amount of chip area.

On the other hand, using DRAM as CPU cache would reduce cost but increase processor waiting time because DRAM has higher latency.

Therefore, SRAM and DRAM are **complementary technologies**. SRAM provides high-speed cache memory, while DRAM provides large-capacity main memory.

Conclusion

SRAM provides **very low latency, very high bandwidth, and high throughput**, making it suitable for CPU cache memory. DRAM provides **higher capacity and lower cost per bit**, making it suitable for main memory.

Thus, the combination of SRAM cache and DRAM main memory creates an efficient memory hierarchy that balances speed, capacity, throughput, and cost.

3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

Compare Virtual Memory with Paging and Cache Memory Mechanisms in Terms of Access Latency and System Throughput

Introduction

Virtual memory and cache memory are two important mechanisms used in computer systems to improve **memory management and performance**.

Cache memory mainly reduces **memory-access latency** by storing frequently used data close to the CPU. Virtual memory with paging provides a **larger logical address space** and allows programs to run efficiently even when all required data cannot remain in physical memory at the same time. Although cache memory focuses mainly on reducing access latency, virtual memory focuses on efficient memory utilization and process execution. Both mechanisms work together to improve **system throughput, CPU utilization, and application performance**.

1. **Cache memory** improves CPU performance by providing faster access to frequently used data and instructions.
2. **Virtual memory** provides a larger logical address space than the available physical memory.
3. Cache memory is mainly managed by **hardware**, while virtual memory is managed by the **Operating System and MMU**.
4. A **cache miss** requires access to a slower level of memory, while a **page fault** may require data to be retrieved from secondary storage.
5. Frequent page faults can reduce system throughput and may cause **thrashing**.
6. Cache and virtual memory serve different purposes but work together to improve overall system performance.

Comparison

Parameter	Cache Memory	Virtual Memory with Paging
Main Purpose	Reduce memory-access latency	Provide a larger virtual address space
Managed By	Hardware / Cache Controller	Operating System and MMU
Unit of Transfer	Cache Line	Page
Main Memory Used	SRAM Cache	DRAM and Secondary Storage

Parameter	Cache Memory	Virtual Memory with Paging
Access Latency	Very low	Higher
Failure Event	Cache Miss	Page Fault
Throughput	Very high	Can decrease during excessive paging
Main Advantage	Faster CPU execution	Larger address space and memory management

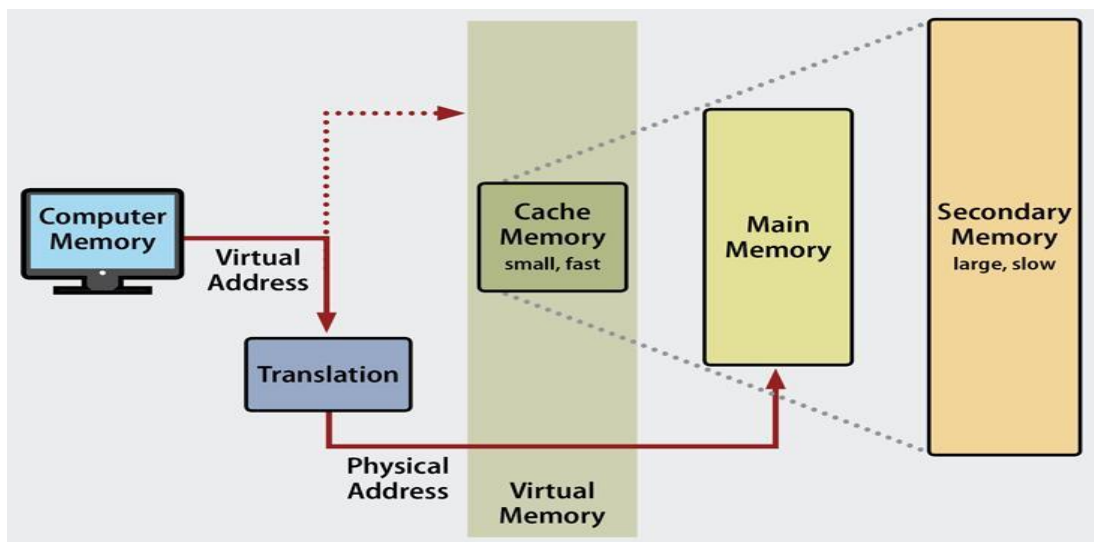


Fig: Comparison of Cache Memory and Virtual Memory with Paging

Access Latency

Cache memory stores frequently used data and instructions close to the processor. When the required data is found in the cache, it is called a **cache hit**, and the CPU can access the data very quickly.

Virtual memory divides memory into fixed-size **pages**. If the required page is already available in physical memory, the program can continue execution normally.

However, if the required page is not available in RAM, a **page fault** occurs. The operating system may need to retrieve the required page from secondary storage, which causes significantly higher access latency.

A typical memory hierarchy is:

CPU → Cache → DRAM → Secondary Storage

System Throughput

Cache memory improves processor throughput because frequently used data can be accessed without repeatedly waiting for slower DRAM.

Virtual memory improves memory utilization and allows larger programs to run. However, frequent page faults can significantly reduce system throughput.

When page faults occur continuously, the system may experience **thrashing**, where a large amount of time is spent moving pages between memory and storage instead of executing programs.

Evidence

Cache access is typically measured in **CPU cycles or nanoseconds**, while a page fault that requires data to be retrieved from secondary storage can take **microseconds or more**, depending on the storage device and system conditions. Therefore, cache hits provide much faster access, while frequent page faults can significantly increase waiting time and reduce overall system throughput.

Critical Analysis

Cache memory and virtual memory serve different purposes:

- **Cache memory** is primarily a **performance mechanism** that reduces memory-access time.
- **Virtual memory with paging** is primarily a **memory-management mechanism** that provides address translation, memory protection, isolation, and efficient memory utilization.

A high cache-hit rate improves processor performance, while a low page-fault rate helps maintain good system throughput.

Therefore, a computer system requires both mechanisms to balance **speed, memory capacity, protection, and efficient resource utilization**.

Conclusion

Cache memory provides **very low latency and high throughput**, making it essential for fast CPU operation.

Virtual memory with paging provides a **larger logical address space, memory protection, and efficient memory management**.

Thus, cache memory mainly improves speed and processor performance, whereas virtual memory improves **capacity, flexibility, and memory management**. Both mechanisms work together to provide efficient overall system performance.

- 4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

Analyze NAND Flash and DRAM with Respect to Latency, Bandwidth, and Suitability

Introduction

DRAM and NAND Flash are important memory technologies used in modern computer systems. **DRAM (Dynamic Random-Access Memory)** is volatile memory mainly used as main memory, while **NAND Flash** is non-volatile memory mainly used in SSDs and other storage devices.

- 1. **DRAM** provides low latency and is suitable for frequently accessed active data.
- 2. **NAND Flash** provides high storage capacity and retains data without power.
- 3. DRAM generally provides much lower access latency than NAND Flash.
- 4. NAND Flash can provide high sequential bandwidth through parallel operation.
- 5. DRAM and NAND Flash work together to balance **speed, capacity, cost, and persistence**.

Comparison

Parameter	DRAM	NAND Flash
Full Form	Dynamic Random-Access Memory	NAND Flash Memory
Volatility	Volatile	Non-volatile
Latency	Very low	Much higher
Bandwidth	High	High, especially sequential
Random Access	Excellent	Relatively slower
Write Performance	Very high / low-latency writes	Slower writes; erase-before-write overhead
Endurance	Very high	Limited compared with DRAM
Density	High	Very high
Cost per Bit	Higher	Lower
Main Application	Main memory	SSD / Storage

Parameter	DRAM	NAND Flash
Data Retention	Requires power	Retains data without power
Capacity	Large	Very large

Latency

DRAM provides **much lower latency** than NAND Flash. It is therefore suitable for active programs and data that are continuously accessed by the CPU.

NAND Flash has significantly higher access latency because flash memory uses program and erase operations. Therefore, NAND Flash is not suitable as a direct replacement for DRAM in conventional main-memory applications.

Bandwidth and Throughput

DRAM provides high bandwidth for transferring data between the CPU and main memory. Modern systems can increase DRAM bandwidth using **multiple memory channels, memory banks, and burst transfers**.

Modern NAND Flash SSDs can also achieve high **sequential bandwidth** because multiple NAND chips and channels can operate in parallel.

However, **bandwidth and latency are different characteristics**. NAND Flash can provide high sequential bandwidth while still having much higher random-access latency than DRAM.

Suitability in Memory Hierarchy

The typical memory hierarchy is:

CPU → Cache → DRAM → NAND Flash

- **Cache:** Very fast access to frequently used data.
- **DRAM:** Stores active programs and working data.
- **NAND Flash:** Stores the operating system, applications, documents, videos, and other persistent data.

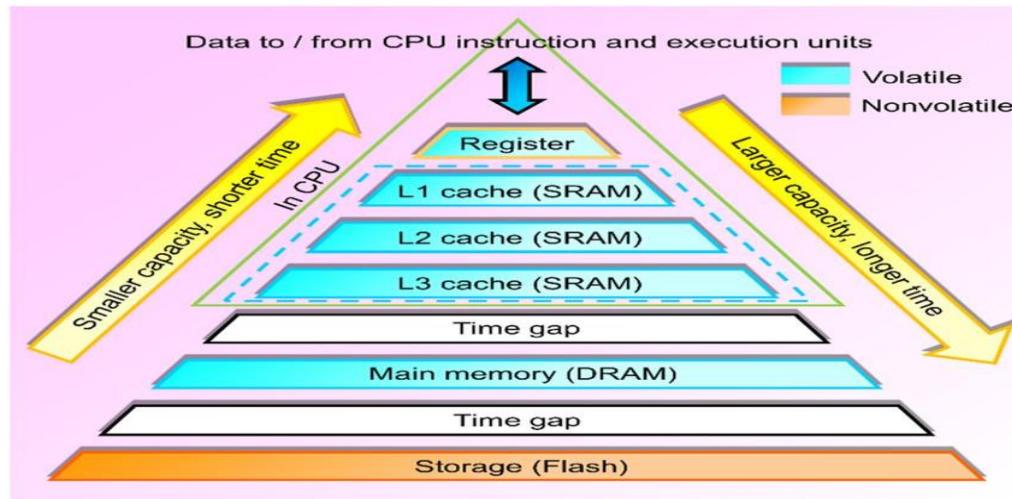


Fig: Computer Memory Hierarchy – Cache, DRAM, and Flash Storage

Evidence

DRAM provides **nanosecond-scale access latency**, making it suitable for frequently accessed active data. NAND Flash has much higher latency, particularly for random access, but modern SSDs can achieve **high sequential bandwidth** by operating multiple NAND chips and channels in parallel. This demonstrates why DRAM is suitable for main memory while NAND Flash is better suited for high-capacity, non-volatile storage.

Critical Analysis

DRAM is better when **low latency and frequent random access** are required.

NAND Flash is better when **large capacity, persistence, and lower cost per bit** are important.

Replacing DRAM with NAND Flash would reduce system performance because of its higher latency.

Replacing NAND Flash with DRAM would provide faster access but would increase cost and would not provide non-volatile storage.

Therefore, both technologies are **complementary rather than direct replacements**.

Conclusion

DRAM provides **low latency, high bandwidth, and fast access**, making it suitable for main memory.

NAND Flash provides **high capacity, non-volatility, and lower cost per bit**, making it suitable for persistent storage.

Thus, modern computer systems use both technologies to achieve a balance between performance, capacity, cost, and data persistence.

5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Compare MRAM and RRAM in Terms of Access Time, Throughput, and Energy Efficiency

Introduction

MRAM (Magnetoresistive Random-Access Memory) and **RRAM (Resistive Random-Access Memory)** are emerging **non-volatile memory** technologies. Unlike conventional volatile memory, both can retain stored data even when power is removed.

1. **MRAM** stores data using magnetic states.
2. **RRAM** stores data by changing the resistance of a material.
3. Both technologies provide non-volatile data storage.
4. MRAM is mainly attractive for **fast access and high endurance**.
5. RRAM has strong potential for **high density and parallel processing**.

Comparison

Parameter	MRAM	RRAM
Full Form	Magnetoresistive RAM	Resistive RAM
Volatility	Non-volatile	Non-volatile
Access Time	Very low	Low to moderate
Throughput	High	High potential
Standby Energy	Very low	Very low
Endurance	Generally very high	Technology-dependent
Density	High	Very high potential
Main Advantage	Speed and endurance	Density and parallelism

Parameter	MRAM	RRAM
Main Limitation	Write energy/current	Variability and endurance
Data Retention	Yes	Yes
Suitable Application	Embedded systems, cache/storage	High-density memory, AI/compute-in-memory

Access Time

MRAM provides fast read and write access while retaining data when power is removed. This makes it suitable for applications where both speed and non-volatility are required.

RRAM stores information by changing the resistance of a material. Its simple memory-cell structure provides potential for **fast switching and high-density memory**.

Therefore, MRAM is generally attractive when **low access time and high endurance** are important.

Throughput

Both MRAM and RRAM have the potential to provide high throughput.

MRAM is useful for applications requiring frequent memory operations and high endurance.

RRAM has particular potential for **parallel processing and compute-in-memory**, where multiple memory cells can participate in operations simultaneously.

Therefore, RRAM can be particularly useful for workloads that benefit from high levels of parallelism.

Energy Efficiency

Both MRAM and RRAM have very **low standby energy consumption** because they are non-volatile and do not require continuous refresh.

MRAM can provide good standby energy efficiency, although some MRAM designs require relatively high current during write operations.

RRAM can also provide low standby power, but its energy consumption depends on the specific resistance-switching mechanism and device design.

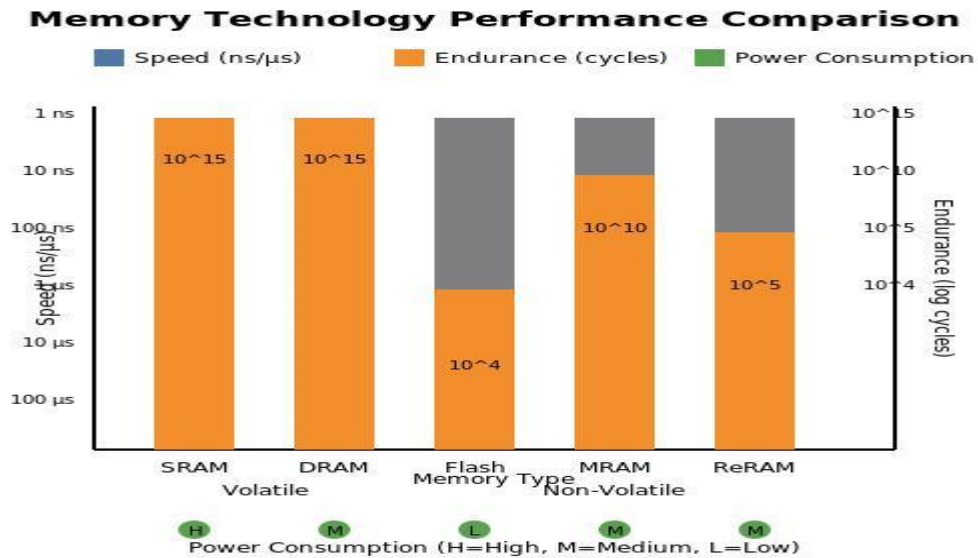


Fig: Energy Efficiency Comparison of MRAM and RRAM

Evidence

MRAM is generally attractive for **fast access, high endurance, and non-volatile operation**, while RRAM has strong potential for **high density, low standby power, and compute-in-memory applications**. RRAM can also support parallel operations in suitable architectures. However, exact access time, throughput, write energy, and endurance depend on the specific device technology and implementation.

Critical Analysis

Neither MRAM nor RRAM is universally better. The choice depends on the application requirements.

MRAM is more suitable when:

- Fast access is required.
- High endurance is important.
- Non-volatility is required.
- Frequent read/write operations are performed.

RRAM is more suitable when:

- High memory density is required.

- Parallel processing is important.
- Low standby power is desired.
- Compute-in-memory applications are required.

The main trade-off is between **speed, endurance, density, energy consumption, and manufacturing complexity.**

Conclusion

MRAM provides an attractive combination of fast access, high endurance, and non-volatility, making it suitable for high-performance and embedded applications.

RRAM offers strong potential for **high density, low standby energy, and parallel computing.**

Therefore, both MRAM and RRAM are promising technologies for future memory systems, with the final choice depending on the required access time, throughput, energy efficiency, density, and endurance.

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Signature of the student:

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